

Course Syllabus

SIO30 – The Oceans 2022 – Summer Session I (06/27 – 07/30)

Instructor: Tiago Carrilho Biló (tcarrilhobilo@ucsd.edu)

Teaching Assistant: Michelle Guraieb (mguraieb@ucsd.edu)

Classes: TuTh 2:00-4:50 pm

Discussion Sessions: Fridays 2:00-3:50 pm

Suggested Reading: Garrison & Ellis, Oceanography: An Invitation to Marine Science, 9th (or older) Edition

This is an introduction to Oceanography course intended for both science and non-science students. You are expected to learn about the general topics listed below; these will be covered during lectures and will be discussed further during discussion sessions. Your grade will be based on weekly homework tasks (four in total) and a final exam. Exam questions will be taken primarily from lecture materials and less so from the suggested reading and discussion sessions.

Grades (Maximum 10 pts): -> Homework – 60% (average of the four homework tasks);

-> Final Exam – 40%

-> Participation: for those with active participation in class and discussion sessions will be awarded with extra 2 pts (questions, observations, and active role on discussions).

Tentative schedule

Week 1 – Marine Geology

Jun 28th Tu 2:00 pm Course Introduction and Origin of the Earth and Oceans

Jun 28th Tu 3:00 pm Earth Structure and Plate Tectonics

Jun 28th Tu 4:00 pm Plate Tectonics – continuation

Jun 30th Th 2:00 pm Ocean Basins & Margins (ocean's geography)

Jun 30th Th 3:00 pm Marine sediments

Jun 30th Th 4:00 pm Marine sediments – continuation

Jul 1st Fr 2:00 pm – Marine Geology Discussion

Week 2 – Marine Chemistry

Jul 5th Tu 2:00 pm (Homework 1 due date) Ocean's water structure

Jul 5th Tu 3:00 pm Water components I

Jul 5th Tu 4:00 pm Water components II

Jul 7th Th 2:00 pm Gases in Seawater and Ocean's PH

Jul 7th Th 3:00 pm Oxygen, Nitrogen, and Carbon in the Ocean (Guest Lecturer Monica)

Jul 7th Th 4:00 pm Marine Pollution

Jul 8th Fr 2:00 pm – Marine Chemistry Discussion

Week 3 – Physical Oceanography

Jul 12th Tu 2:00 pm (Homework 2 due date) Physical properties of the ocean:
temperature, salinity, and density

Jul 12th Tu 3:00 pm Forces acting on the ocean

Jul 12th Tu 4:00 pm Forces in action

Jul 14th Th 2:00 pm Global circulation

Jul 14th Th 3:00 pm Ocean Circulation – Continuation (Coastal Oceans – Guest Lecturer
Robert Sanchez)

Jul 14th Th 4:00 pm Waves and Tides

Jul 15th Fr 2:00 pm – Physical Oceanography Discussion

Week 4 – Biological Oceanography

Jul 19th Tu 2:00 pm (Homework 3 due date) Overview, terminology, biomes of the sea (Guest Lecturer Michelle Guraieb).

Jul 19th Tu 3:00 pm Primary producers

Jul 19th Tu 4:00 pm Marine animals

Jul 21st Th 2:00 pm Interactions: nutrient cycles and the circle of life

Jul 21st Th 3:00 pm Plankton, light, and ocean ecosystems

Jul 21st Th 4:00 pm Climate change and sea level rise (Guest Lecturer Margeret Lindeman)

Jul 22nd Fr 2:00 pm – Biological Oceanography Discussion

Week 5 – Review and Final Exam

Jul 26th (Homework 4 due date) – Course Review

Jul 28th Final Exam (closed book)

Jul 29th Final Exam discussion (TBD)